

Varshap Walia

Backend & AI Systems Engineer with experience building distributed systems, scalable platforms, and production-grade LLM workflows.

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TECHNICAL SKILLS

Languages: Go, Java, Python, C#, JavaScript, SQL

Backend & Distributed Systems: Microservices, Kafka, Redis, REST APIs, gRPC, JWT, OAuth, OIDC, SAML, SCIM, Event-driven Architecture

Frameworks and Libraries: .NET / ASP.NET, Spring Boot, FastAPI, Node.js, React

AI/ML: LLM Agents, RAG, LangGraph, LlamaIndex, FAISS, Vector Databases, Prompt Engineering, PyTorch

Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Helm, GitLab CI/CD, GitHub Actions, Kong Gateway

Databases & Observability: PostgreSQL, MongoDB, MySQL, Datadog, New Relic, Sentry

EXPERIENCE

Software Engineer

Jan. 2025 – Present

Thales Group

Ottawa, ON

- Built autonomous **CI/CD failure recovery and test self-healing system** using **LLM agents** with tool execution and browser control, automatically resolving **~35–40%** of recurring flaky test failures.
- Reduced p95 latency in critical **IAM workflows** from **~800ms** to under **120ms (~6× improvement)** through query optimization, indexing, and distributed caching.
- Built **RAG-based incident analysis platform** integrating Jira, logs, traces, and metrics using **FAISS retrieval**, reducing MTTR by an estimated **20–30%**.
- Led **LLM-based .NET 5 → .NET 8 migration framework** for core feature services with multi-pass validation and feedback loops, achieving **~70%** to **~82%** automation.
- Developed and scaled **IAM microservices** in .NET and Java supporting **SAML, OAuth, and OIDC** across multi-tenant environments serving **1M+ users**.
- Built distributed **load testing framework** using Kubernetes and Locust, simulating **100K+ concurrent users** and identifying bottlenecks before production deployment.
- Optimized **LLM inference pipelines** using retrieval, context pruning, and multi-layer caching, reducing token consumption by up to **3–4×**.

Software Development Engineer II

Feb. 2022 – Aug. 2022

Vahak

Bengaluru, IN

- Redesigned geolocation platform built on **Go, Elasticsearch, and OpenStreetMap data**, expanding points-of-interest coverage by **4×** while improving data integrity and matching accuracy.
- Improved search relevance by **~89%** through relevance tuning and query analysis, increasing conversion.
- Built **Kafka-based pipelines** processing **~2M events/day** into MongoDB and S3 for analytics and ML workflows.

Software Engineer

Apr. 2019 – Jan. 2022

Pinkblue

Bengaluru, IN

- Identified manual handoff steps inflating delivery error rates; redesigned logistics workflow with **automated validation gates**, reducing errors **~17%** and saving an estimated **\$187K annually**.
- Architected and built **POS and logistics platform** across **100+ hospitals and 450+ pharmacies**, unifying inventory, order processing, and fulfillment as the first system to span both verticals.
- Led **GCP to AWS migration**, improving observability with New Relic and Sentry, reducing incidents by **~23%**.
- Built **real-time pricing engine** using market and vendor data feeds, improving order conversion by **~15%**.

Backend Engineer

Nov. 2017 – Mar. 2019

Inclov Technologies

Gurgaon, IN

- Built encrypted messaging and push notification systems from scratch; improved delivery reliability from **72% → 94%** and increased engagement by **~41%** over six months post-launch.
- Implemented **ML-based content moderation pipeline** using OpenCV and Scikit-learn, reducing manual moderation workload by **~27%** and enabling headcount-free scaling.
- Redesigned onboarding flows for **WCAG 2.1 accessibility compliance**, improving activation rates by **20–25%** based on funnel tracking before and after rollout.

EDUCATION

Concordia University

Master of Applied Computer Science, Major in AI

Montreal, QC

Sept. 2022 – Dec. 2024